

Approach A (Direct LoRA): Each image gets 1 training pair — the image plus the full classification prompt as input, and the correct JSON label as output. This produces N training examples.

Approach B (Augmented LoRA): Each image gets 4 training pairs — 3 CoT pairs (image + prompt without JSON instruction → GPT-4.1 description + JSON appended by code) and 1 direct pair (same as Approach A). This produces $4N$ training examples from the same N images. The JSON is never generated by the frontier model; it is appended programmatically to guarantee format consistency.

Figure 1 illustrates both training approaches side by side.

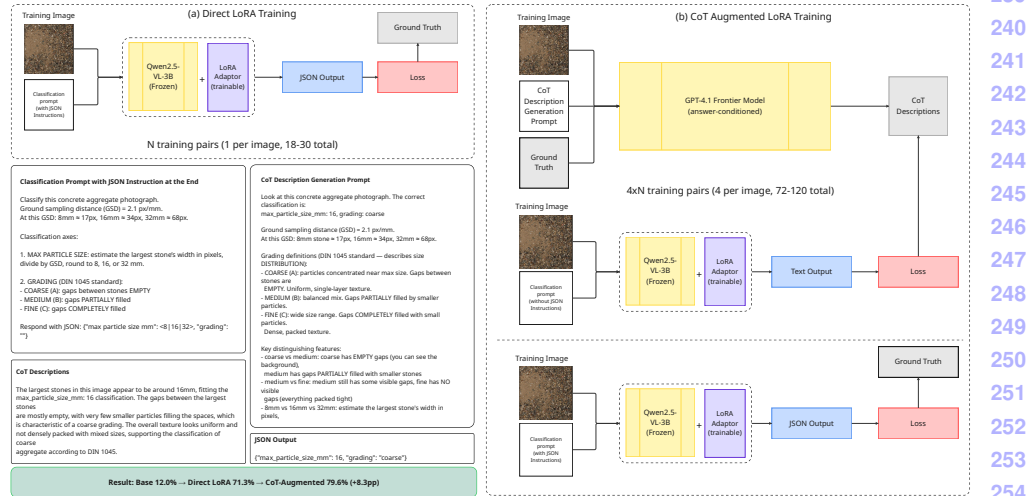


Figure 1: Method overview. (a) Direct LoRA: the model learns image→JSON mappings from N training pairs. (b) CoT-augmented LoRA: GPT-4.1 receives each image with the correct label and generates justified visual descriptions; combined with direct pairs, this yields $4N$ training examples from the same N images.

3.5 LoRA Fine-Tuning

Both approaches use identical LoRA [8] hyperparameters applied to the Qwen2.5-VL-3B-Instruct model [14]:

During training, only the assistant response tokens contribute to the loss. All user-prompt and image tokens are masked with label value -100 .

At evaluation, both models receive the full classification prompt (with JSON instruction) and generate a response with temperature 0.1. The JSON is parsed from the output, extracting it from within any surrounding description text if present.

3.6 Why Answer-Conditioned CoT Distillation Helps

The augmented approach outperforms Direct LoRA for two reasons: